ITA0608 – Machine Learning Lab Practicals

EXP 09:

Compare Linear and Polynomial Regression using Python

Aim:

To implement and compare Linear Regression and Polynomial Regression models using Python and evaluate their performance on a dataset.

Algorithm:

- Load the dataset and split it into training and testing datasets.
- Train a Linear Regression model and make predictions.
- Transform the features into polynomial features and train a Polynomial Regression model.
- Predict the output using both models.
- Compare the models using Mean Squared Error (MSE) and R^2 Score.

Program:

```
from sklearn.datasets import load_diabetes

from sklearn.model_selection import train_test_split

from sklearn.linear_model import LinearRegression

from sklearn.preprocessing import PolynomialFeatures

from sklearn.metrics import mean_squared_error, r2_score

diabetes = load_diabetes()

X = diabetes.data      y = diabetes.target

X_train, X_test, y_train, y_test = train_test_split(

   X, y, test_size=0.3, random_state=42 )

linear_model = LinearRegression()

linear_model.fit(X_train, y_train)

linear_pred = linear_model.predict(X_test)

linear_mse = mean_squared_error(y_test, linear_pred)

linear_r2 = r2_score(y_test, linear_pred)

poly = PolynomialFeatures(degree=2)

X_train_poly = poly.fit_transform(X_train)

X_test_poly = poly.transform(X_test)
```

```
poly_model = LinearRegression()
poly_model.fit(X_train_poly, y_train)
poly_pred = poly_model.predict(X_test_poly)
poly_mse = mean_squared_error(y_test, poly_pred)
poly_r2 = r2_score(y_test, poly_pred)
print("Linear Regression Results")
print("--------------------------")
print("Mean Squared Error:", round(linear_mse, 2))
print("R² Score:", round(linear_r2, 2))
print("\nPolynomial Regression Results")
print("------------------------------")
print("Mean Squared Error:", round(poly_mse, 2))
print("R² Score:", round(poly_r2, 2))
new_sample = [X[0]]
linear_prediction = linear_model.predict(new_sample)
poly_prediction = poly_model.predict(poly.transform(new_sample))
print("\nPrediction for New Sample")
print("Linear Regression :", round(linear_prediction[0], 2))
print("Polynomial Regression :", round(poly_prediction[0], 2))
```

Output:

```
... Linear Regression Results
    --------------------------
    Mean Squared Error: 2821.75
    R² Score: 0.48

    Polynomial Regression Results
    ------------------------------
    Mean Squared Error: 3168.93
    R² Score: 0.41

    Prediction for New Sample
    Linear Regression : 209.69
    Polynomial Regression : 216.86
```

Result:

The Linear Regression and Polynomial Regression models were successfully implemented and compared using Python